

Here-doc / here-string

Sometimes you need to feed a command several lines of input without creating a temp file

Multi-line stdin vs one string

- Less-than less-than EOF starts a here-doc
- Leave the delimiter unquoted if you want variables expanded
- Three less-thans with a string is a here-string, one chunk of text as stdin
- Prefer these over awkward echo pipes when the reader must run in the current shell

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

Here-doc / here-string

Feed multi-line stdin with <<EOF ... EOF, contrast quoted delimiters, and use <<< for a one-line here-string.

Starter example: cat <<EOF with three lines — the body is stdin, not a file. Toggle << 'EOF' to freeze \$HOME, or switch to read name <<< "alice".

Load starter example

Challenges 0 / 22 cleared

Quiz: here-doc: <<EOF feeds data to the command's? Answer: stdin

Answer

Show hint

Check

Next

Idle

Quiz: here-doc

Run cat

Quiz: delimiter

grep two

wc lines

Quiz: expand

Expand HOME

Quoted literal

Quiz: 'EOF'

Custom END

Quiz: <<<

read <<<

cat <<<

Quiz: why

Quiz: pipe read

Three lines

Script shape

HOME value

wc here-string

Quiz: closing

grep ERROR

Quiz: both

Build input

<< here-doc

<<< here-string

Lab env: HOME=/home/lab · USER=lab · PWD=/home/lab/chip

Command

Script · stdin · stdout

What the shell sees

```
cat <<EOF
line one
line two
```

Here-doc lab

Real shell practice

```
wsl - module13-here-doc-lab/examples/here_doc

# Track A - real Unix
# real shell session (Track A)

$ chmod u+x read_stdin.sh

$ cat <<EOF
line one
line two
EOF
line one
line two

$ grep two <<EOF
one
two
three
EOF
two

$ ./read_stdin.sh <<EOF
fed via here-doc
EOF
Reading from stdin (use <<EOF ... EOF to feed input):
  Got: fed via here-doc

$ grep hi <<< "hi from here-string"
hi from here-string
```

Real shell — here-doc and here-string

Real shell practice — try these

```
# chmod u+x read_stdin.sh — make the stdin reader executable
chmod u+x read_stdin.sh
```

```
# cat <<EOF ... EOF — here-doc: multi-line body becomes cat's stdin
cat <<EOF
```

```
line one
```

```
line two
```

```
EOF
```

```
# grep two <<EOF ... EOF — search inside a here-doc body
grep two <<EOF
```

```
one
```

Pitfalls to watch

- The closing delimiter must be alone on its line, no spaces, or the here-doc never ends
- Unquoted EOF expands variables; quoted EOF does not, pick deliberately
- And remember

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, finish a few challenges after the starter
- On the real shell, practice a here-doc into cat and the reader script, plus a here-string
- When you are ready, take the short quiz, then continue to shell history and reverse-search